

TAIF, Saudi Arabia

WMO: 410360

Lat: 21.483N

Lon: 40.544E

Elev: 1478

StdP: 84.79

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.9	8.3	-11.8	1.6	27.4	-8.5	2.2	24.1	10.7	20.7	10.0	21.0	1.7	180	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.0	37.2	16.7	36.5	16.7	35.9	16.6	21.8	30.5	20.7	30.1	19.8	30.0	4.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	16.6	27.4	17.9	15.4	26.0	16.8	14.4	24.7	71.9	30.0	67.2	30.2	63.9	30.0	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.4	8.6	DB	3.5	39.1	1.3	1.6	2.5	40.2	1.8	41.2	1.0	42.1	0.1	43.2	
			WB	0.1	24.4	1.1	2.9	-0.7	26.5	-1.3	28.1	-1.9	29.7	-2.7	31.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.6	16.1	18.1	20.4	23.3	26.6	29.6	29.7	30.1	28.2	23.7	19.7	17.0
	DBStd	5.39	2.46	2.41	2.52	2.08	2.04	1.57	1.86	1.46	1.56	1.76	1.90	2.04
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	172	75	29	11	1	0	0	0	0	0	0	8	48
	CDD10.0	4950	189	228	321	398	514	588	610	622	546	424	291	218
	CDD18.3	2081	6	24	74	149	256	338	352	364	296	166	49	7
	CDH23.3	25557	84	263	694	1435	2968	4641	4633	4791	3721	1764	457	106
	CDH26.7	12624	6	34	160	530	1490	2612	2482	2587	1958	691	70	4
Wind	WSAvg	3.8	3.1	3.6	4.0	3.8	3.5	4.4	5.6	5.1	3.5	2.9	2.8	2.9
Precipitation	PrecAvg	151	9	1	10	28	31	3	2	15	10	16	20	6
	PrecMax	299	38	15	53	113	143	21	11	135	55	118	235	34
	PrecMin	66	0	0	0	1	1	0	0	0	0	0	0	0
	PrecStd	64	10	3	14	29	35	6	3	29	14	25	45	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.2	29.2	31.2	33.2	36.2	38.1	38.2	38.1	37.1	33.2	29.9	27.1
		MCWB	13.9	14.4	15.1	15.6	16.4	16.7	16.6	16.8	16.8	15.2	14.9	14.1
	2%	DB	25.2	27.8	29.9	32.0	35.2	37.1	37.1	37.1	35.9	32.1	28.2	25.7
		MCWB	13.6	14.5	14.8	16.1	16.2	16.3	16.6	17.1	16.3	14.9	15.2	14.0
	5%	DB	24.0	26.2	28.3	31.0	34.2	36.1	36.0	36.1	35.0	31.2	27.2	24.4
		MCWB	13.8	14.6	14.9	16.3	16.1	16.2	16.7	17.4	16.2	14.6	14.9	14.2
	10%	DB	22.8	25.0	27.1	29.9	33.1	35.2	35.0	35.1	34.0	30.2	26.1	23.2
		MCWB	13.6	14.4	14.9	16.1	16.2	16.1	16.5	17.5	16.1	14.5	14.7	14.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	18.0	19.0	20.4	21.8	22.0	22.4	24.0	22.0	21.2	20.4	17.3
		MCDB	23.0	24.4	25.8	28.3	30.2	32.2	31.7	32.2	29.8	26.1	24.9	23.0
	2%	WB	16.0	16.8	17.9	19.0	19.7	20.1	20.6	21.8	20.5	19.1	18.1	16.4
		MCDB	22.0	23.7	25.0	27.0	29.2	32.1	31.1	31.1	28.7	24.0	23.4	22.4
	5%	WB	15.3	16.1	17.0	18.2	18.9	19.0	19.5	20.7	19.5	18.2	17.2	15.8
		MCDB	21.1	23.2	24.7	26.4	28.8	31.7	31.3	30.8	28.8	24.2	23.0	21.8
	10%	WB	14.6	15.4	16.2	17.5	18.1	17.9	18.4	19.7	18.5	17.4	16.6	15.2
		MCDB	20.5	22.6	24.2	26.0	28.2	31.6	31.2	31.0	29.1	24.4	22.7	21.1
Mean Daily Temperature Range	5% DB	MDBR	13.1	13.6	13.6	13.3	13.3	12.1	10.6	11.0	12.8	13.7	13.0	13.0
		MCDBR	14.5	14.6	14.5	14.2	14.0	12.7	11.4	11.9	13.1	14.2	14.5	14.3
		MCWBR	7.0	6.9	6.4	5.5	5.4	5.5	5.5	5.8	5.9	6.3	6.3	6.7
	5% WB	MCDBR	12.8	13.7	13.1	13.0	12.9	12.2	10.7	11.4	13.0	13.6	13.0	12.7
		MCWBR	6.9	7.1	6.4	5.8	5.5	6.2	6.3	6.5	6.7	6.8	6.1	6.4
Clear-Sky Solar Irradiance	taub		0.344	0.377	0.411	0.457	0.505	0.517	0.540	0.565	0.510	0.428	0.380	0.352
	taud		2.310	2.187	2.070	1.963	1.828	1.786	1.764	1.708	1.802	2.009	2.180	2.276
	Ebn at Noon		923	911	893	853	806	789	772	755	791	845	873	899
	Edh at Noon		118	142	167	189	214	222	227	240	215	168	134	119
All-Sky Solar Radiation	RadAvg		4.82	5.72	6.59	6.99	7.47	7.95	7.55	7.05	6.70	6.11	5.05	4.59
	RadStd		0.26	0.22	0.27	0.32	0.34	0.21	0.19	0.14	0.18	0.19	0.23	0.21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	-35	+154	+127
Regional (3 neighbors)	+0.48	+0.70	-1.24	N/A	N/A	N/A	N/A	N/A	+188	+172

Nomenclature: See separate page